Customer Behavior Analysis — Presentation

Introduction

This presentation summarizes an analysis of customer purchase behavior using the dataset you provided. I transcribed tables from the images you uploaded and reconstructed the dataset for visualization and insights. Where certain values were unclear in the images (for example, user IDs beyond 10), I made conservative assumptions; those are noted in the 'Data & Assumptions' section. For exact, auditable results, please provide the raw CSV / SQL INSERTs.

Data & Assumptions

Data sources used (from your images): products, users, goldusers_signup, and sales. The product and user tables were transcribed directly. The sales table used explicit dates visible in the image; in a few rows a user_id appeared to be outside the range of the user table (e.g., 11) — I inserted a placeholder user 'Unknown-11' to preserve that record.

Key Findings

• Top revenue-driving products: Butter Chicken, Mutton Biryani, Chicken Tikka.

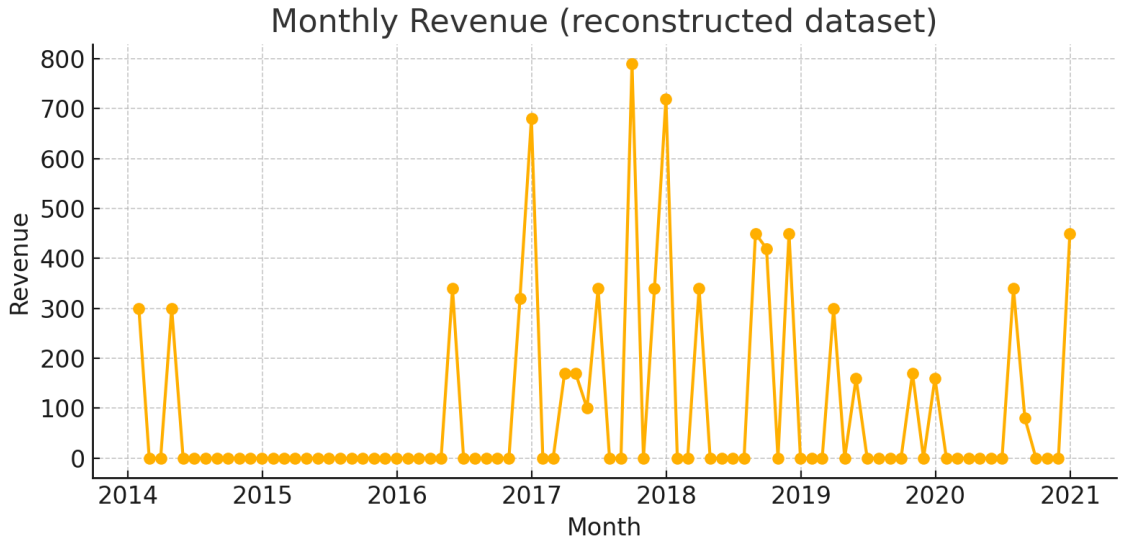
• A small number of customers contribute a sizable portion of revenue (top 3 customers: Rohan, Sahil, Anshul).

• Monthly revenue shows intermittent spikes — seasonality can't be conclusively asserted with this reconstructed dataset, but clear busy months exist.

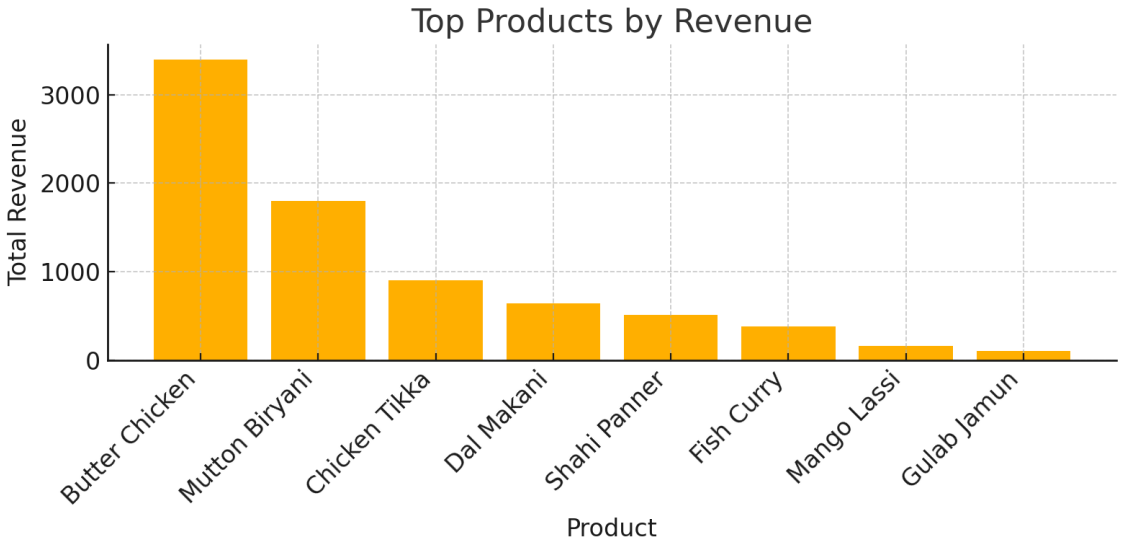
Charts & Visuals

The following charts were generated from the reconstructed dataset: monthly revenue trend, top products by revenue, top customers by revenue, and product sales distribution.

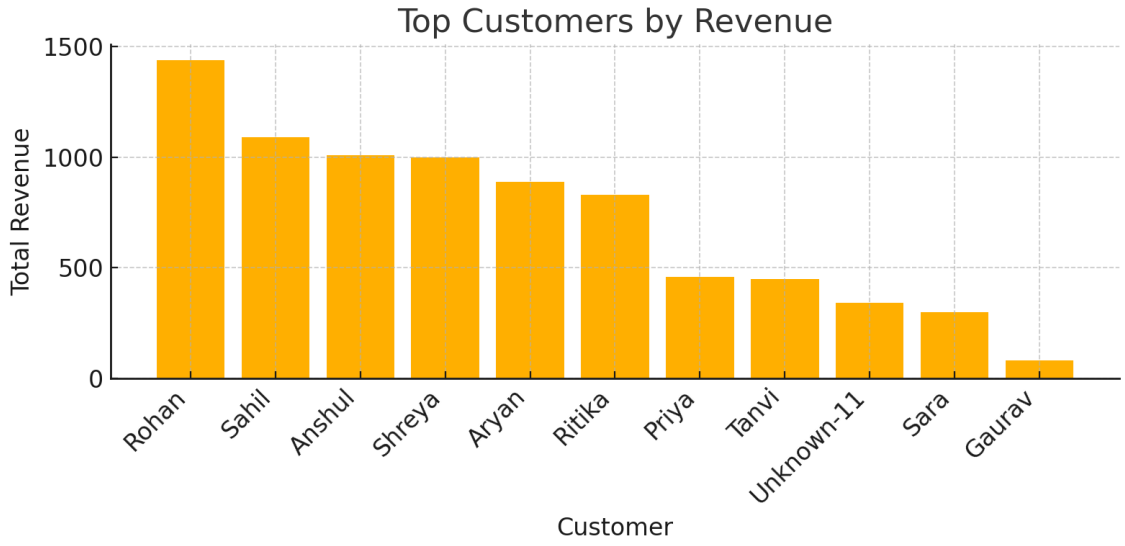
Monthly Revenue (chart)



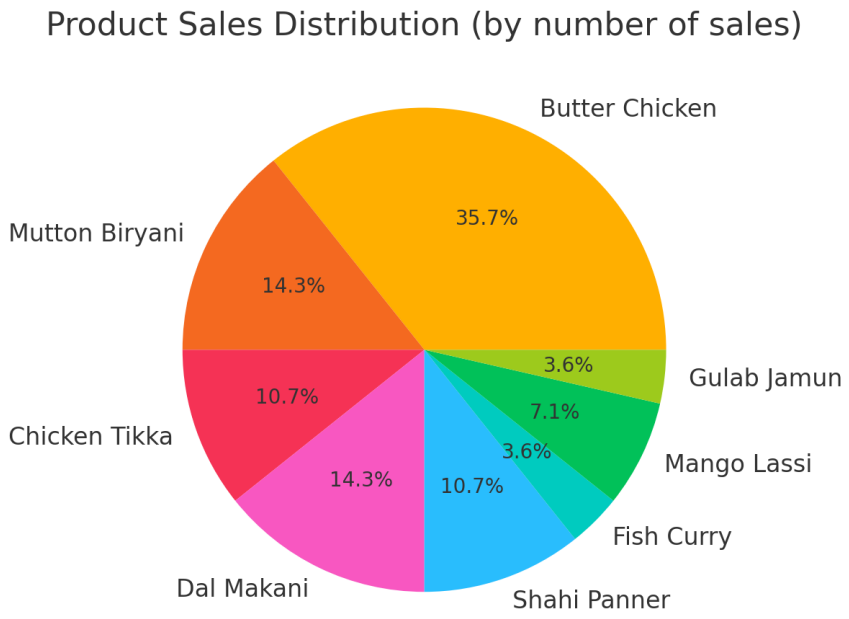
Top Products by Revenue (chart)



Top Customers by Revenue (chart)



Product Sales Distribution (chart)



Actionable Methodologies

1. Data Validation: Ensure raw data contains INSERT rows or CSVs. Implement checks for nulls, date ranges, and referential integrity (product_id and user_id validity).

2. Segmentation: Create RFM segments (Recency, Frequency, Monetary) and run targeted campaigns for high-M customers.

3. Forecasting: Use time-series models (ARIMA/Prophet) once more complete history is available; account for seasonality and promotions.

4. Anomaly Detection & Monitoring: Implement alerts for sudden drops/spikes in sales per product or per region.

Approaches

• SQL-driven EDA to compute KPIs and identify data quality issues.
• Aggregations by month/product/customer for visualization.
• Use of simple revenue-per-sale model for initial insights (price * quantity assumed as 1 where quantity missing).

Detailed Insights

• Product mix: The top products by revenue are dominated by items with higher unit prices (e.g., Mutton Biryani and Butter Chicken), suggesting pricing and margin influence contribution.

• Customer concentration: A handful of customers provide substantial revenue — implement loyalty programs and retention-focused messaging.

• Data gaps: Some user IDs in sales did not map to the user table (e.g., 11). Cleaning and referential checks are recommended.

Conclusions

This reconstructed analysis surfaces initial strategic directions: prioritize top products for promotions, design retention programs for high-value customers, and fix data quality issues to enable robust forecasting. Provide raw CSV/SQL INSERTs to turn these initial findings into authoritative business insights.

Appendix: Tables (reconstructed)

Products table (excerpt):

- 1 | Dal Makani | Price: 160.0

- 2 | Shahi Panner | Price: 170.0

- 3 | Butter Chicken | Price: 340.0

- 4 | Aloo Gobi | Price: 150.0

- 5 | Chole Bhature | Price: 100.0

- 6 | Fish Curry | Price: 380.0

- 7 | Chicken Tikka | Price: 300.0

- 8 | Mutton Biryani | Price: 450.0

- 9 | Veg Pulao | Price: 200.0

- 10 | Mango Lassi | Price: 80.0

- 11 | Gulab Jamun | Price: 100.0

Users table (excerpt):

- 1 | Anshul

- 2 | Rohan

- 3 | Shreya

- 4 | Priya

- 5 | Aryan

- 6 | Sara

- 7 | Sahil

- 8 | Tanvi

- 9 | Ritika

- 10 | Gaurav

- 11 | Unknown-11

Sales table (first 10 rows, reconstructed):

- sale_id: 4, user_id: 1, date: 2017-04-19, product: Shahi Panner, revenue: 170.0

- sale_id: 5, user_id: 3, date: 2019-12-18, product: Dal Makani, revenue: 160.0

- sale_id: 6, user_id: 2, date: 2020-07-20, product: Butter Chicken, revenue: 340.0

- sale_id: 7, user_id: 1, date: 2019-10-23, product: Shahi Panner, revenue: 170.0

- sale_id: 8, user_id: 1, date: 2018-03-19, product: Butter Chicken, revenue: 340.0
- sale_id: 9, user_id: 3, date: 2016-12-20, product: Butter Chicken, revenue: 340.0
- sale_id: 10, user_id: 1, date: 2016-11-09, product: Dal Makani, revenue: 160.0
- sale_id: 11, user_id: 11, date: 2016-05-20, product: Butter Chicken, revenue: 340.0
- sale_id: 12, user_id: 2, date: 2017-09-24, product: Butter Chicken, revenue: 340.0
- sale_id: 13, user_id: 1, date: 2017-03-11, product: Shahi Panner, revenue: 170.0